

Interpretable Molecule Generation: Applying Self-Supervised Discovering of Interpretable Features to Molecule Generation

This work is inspired by these two papers:

- MoleGuLAR: Molecule Generation Using Reinforcement Learning with Alternating Rewards. Manan Goel, Shampa Raghunathan, Siddhartha Laghuvarapu and U. Deva Priyakumar*
- Shi, Wenjie, et al. "Self-supervised discovering of interpretable features for reinforcement learning." IEEE Transactions on Pattern Analysis and Machine Intelligence (2020).

The idea is to address the classical black-box problem of the reinforcement learning processes in the field of the molecule generation, adding to the MoleGuLAR pipeline a block that learns how the molecule is been created. This could build an instrument that highlights the most important parts in the molecule structure, that most likely will be the ones the model pays more attention to. To create this block I think that could be adapted the mechanics used in the Shi paper, from the image field to the SMILES (essentially from 2D to 1D).

For a starting point the idea is to implement the models provided in the repo of the first paper (<https://github.com/devalab/MoleGuLAR/tree/master>) and build an interpreter on them adapting the architecture proposed by Shi et al. If this reveals to be unfeasible the Goel et al pipeline could be implemented from scratch before applying the understandability architecture.